Javier Gomez Mijangos

Computer Science & Mathematics Undergraduate · Université Paris-Saclay

📍 France github.com/javpato ✉ [email] ☎ [phone] in [LinkedIn]

TECHNICAL SKILLS

Programming

Python · TypeScript · JavaScript (ES6+) · C / C++ · SQL **intermediate** · Bash

Web & Frameworks

Next.js · React · MDX · Tailwind CSS · HTML5 / CSS3 · Web Workers · WebAssembly

ML & Data

Flow matching & flow diffusion · Generative models · Data science · Image classification · sql.js (SQLite/WASM)

Graphics & Math

Ray tracing · three.js / react-three-fiber · Mafs · KaTeX · ManimGL · mathjs

Engineering practice

Git · GitHub Actions (CI/CD) · Content-Security-Policy hardening · Technical writing

EDUCATION

Double Degree — Computer Science & Mathematics (LDD IM)

Université Paris-Saclay · France

Rigorous dual curriculum: algorithms & data structures, linear algebra, topology & differential calculus, physics.

LANGUAGES

French — fluent

Spanish — fluent

English — professional working

Mandarin Chinese — studying

HIGHLIGHTS

- Trilingual technical author (FR / ES / EN)
- GitHub: Pull Shark ×2, Pair Extraordinaire ×2, YOLO
- Self-directed, ships production-quality work

PROFILE

Computer-science and mathematics undergraduate who builds real, rigorous software end to end — from a WebAssembly SQL engine to generative-AI experiments — and holds it to an uncompromising standard. Self-motivated and detail-obsessed, I take full ownership of what I build, document it thoroughly, and keep raising the bar. I'm applying to **Bending Spoons First Ascent** to test myself against exceptional peers and contribute from day one.

SELECTED PROJECTS

Learning platform — "Formules Vivantes"

github.com/javpato/learning-website

Next.js 14 · TypeScript · MDX · React · WebAssembly · Python (ManimGL)

- Built an **in-browser SQL course** on a real SQLite engine (`sql.js`/WASM) running in a Web Worker with query timeouts and row caps, plus an automated query-validation harness.
- Implemented a from-scratch **SM-2 spaced-repetition** engine and a custom declarative animation / step-presenter framework with programmatic SVG diagrams.
- Interactive 2D/3D math widgets (Mafs + react-three-fiber) over a shared compute engine; ManimGL animation pipeline; **trilingual i18n** (fr/es/en).
- Led a deliberate **framework-free → Next.js migration**, CSP security hardening, and GitHub Actions CI/CD to GitHub Pages.

Flow-diffusion generative AI

in progress

Python · Flow matching · Diffusion models

- Completed multiple courses on **flow matching and flow diffusion**, and started a project applying them to build and train generative models.

Ray tracer

personal project

Rendering · Computer graphics

- From-scratch **ray tracer**: ray–object intersection, shading/lighting, and recursive reflection to render 3D scenes.

Data-Science-Spoon-knives

university project

Python · Jupyter · Data science

- University data-science project (image classification) built in Jupyter with a reusable Python utilities module and reproducible dependency setup.

INTERESTS

Generative AI · Computer graphics · Databases · Mathematics & pedagogy · Open source · Languages